THE DECISION

Where should the green time go?

A stochastic queuing study of a four-approach traffic signal.

4
incoming approaches

12
movement queues

80 s
green per cycle

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PROBLEM DEFINITION

A precise, limited question

What we choose

Four green durations: North, East, South, West.

What stays fixed

96 s cycle, 16 s clearance, protected phase order, two-second service headway.

Our experiment

Initially empty queues; hypothetical Poisson demand; ten-minute operating period.

Objective: minimize expected cumulative queue waiting.

No real intersection was measured. All performance claims are conditional on the scenario model.

MATHEMATICAL MODEL

From arrivals to waiting cost

Arrivals

$$\Delta t = -\ln(1-U) / \lambda$$

LCG uniform → exponential gap → Poisson stream

Queue cost

$$D_T = \Sigma_m \int_0^T Q_m(t)\, dt$$

Area under 12 waiting-queue curves · vehicle-seconds

Turning draws assign each vehicle left, straight, or right; signal phases decide when each dedicated lane may serve.

OR METHOD

Search, then test on fresh traffic

1
Enumerate
All feasible grid timings

2
Train
Same arrivals for every timing

3
Select
Lowest estimated mean waiting

4
Evaluate
New matched arrivals and intervals

Why simulation?
Signal service stops at red and uses fixed two-second headways. The ordinary continuously serving M/M/1 formula is not an exact solution.

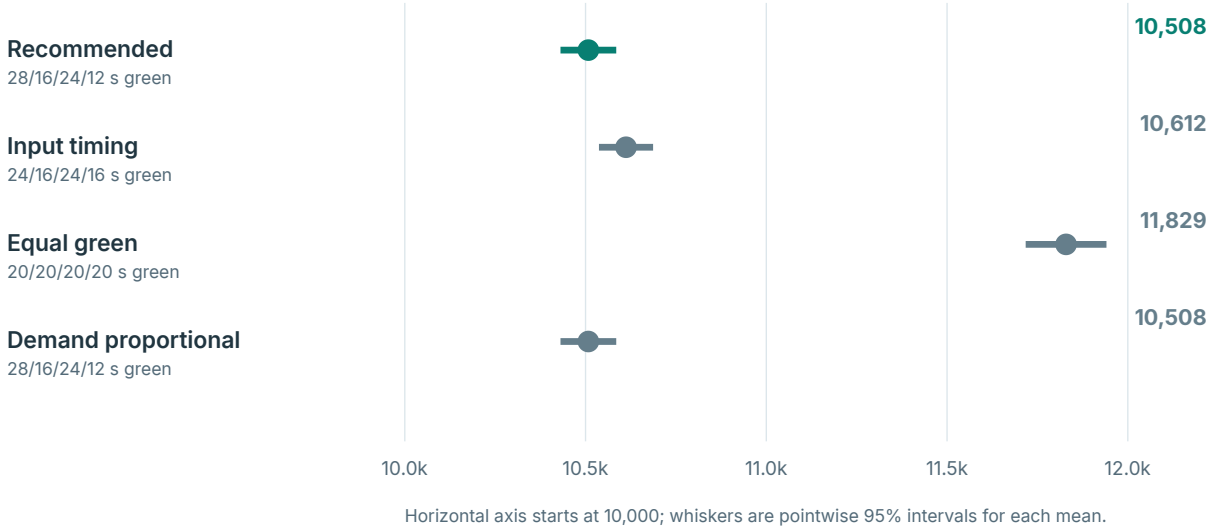
Rust discrete-event core → WebAssembly → SolidJS results and replay.

PRIMARY RESULT

The ten-minute result

Fresh waiting by timing plan

Four-second grid · 1,000 matched fresh scenarios · vehicle-seconds per 10-minute run



Horizontal axis starts at 10,000; whiskers are pointwise 95% intervals for each mean.

Selected green N/E/S/W

28 / 16 / 24 / 12 s

Fresh mean waiting

10,507.8 vehicle-s

vs entered timing, paired saving

104.2 vehicle-s

95% interval: 69.6–138.8 · 1,000 fresh runs

The selected four-second-grid plan equals the demand-proportional baseline. The improvement is modest: ≈0.35 s per expected arrival.

PRECISION STUDY

More precision, small improvement

19,369
feasible one-second timings

6.86 s
cached native search · was 209.02 s

Finer selected timing: **26 / 16 / 24 / 14 s**.

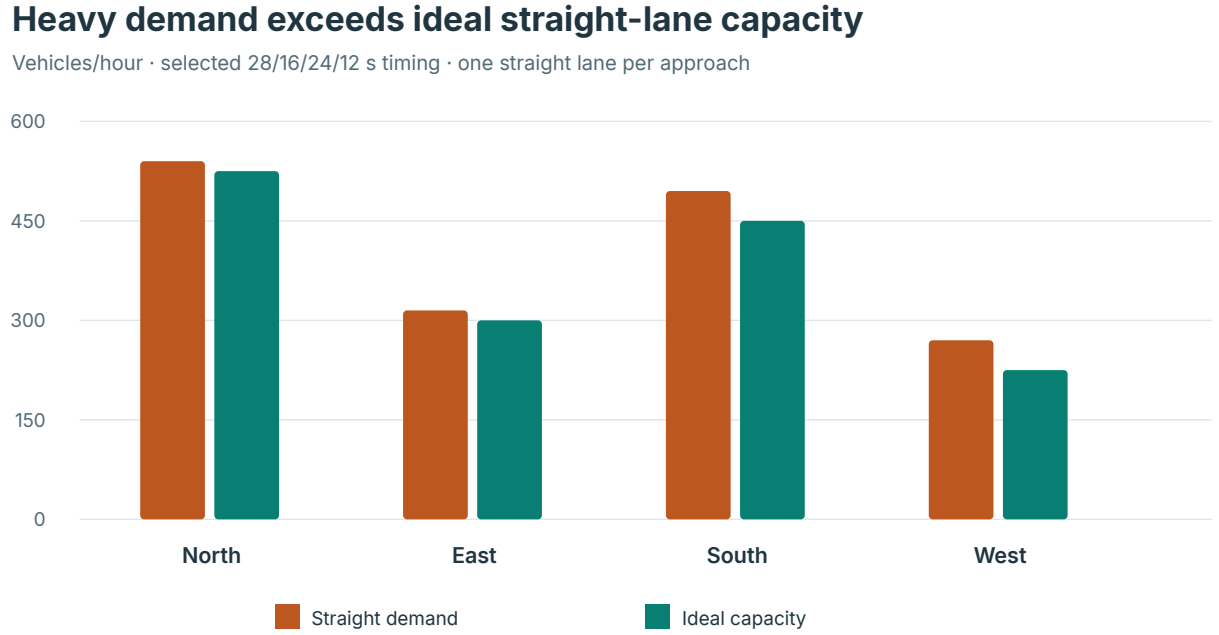
On 5,000 matched fresh runs it saves **44.8 vehicle-seconds** over the four-second-grid plan.

That is only 0.42% of the coarse plan's mean.

Approach caching made this search 30.5× faster and retained the selected plan. 95% paired interval: 34.8–54.8. The small timing gain does not prove global optimality.

CAPACITY STRESS TEST

Demand can outrun capacity



Heavy scenario

2,700 veh/h

Minimum green required by straight streams

86.4 s

Green available

80 s

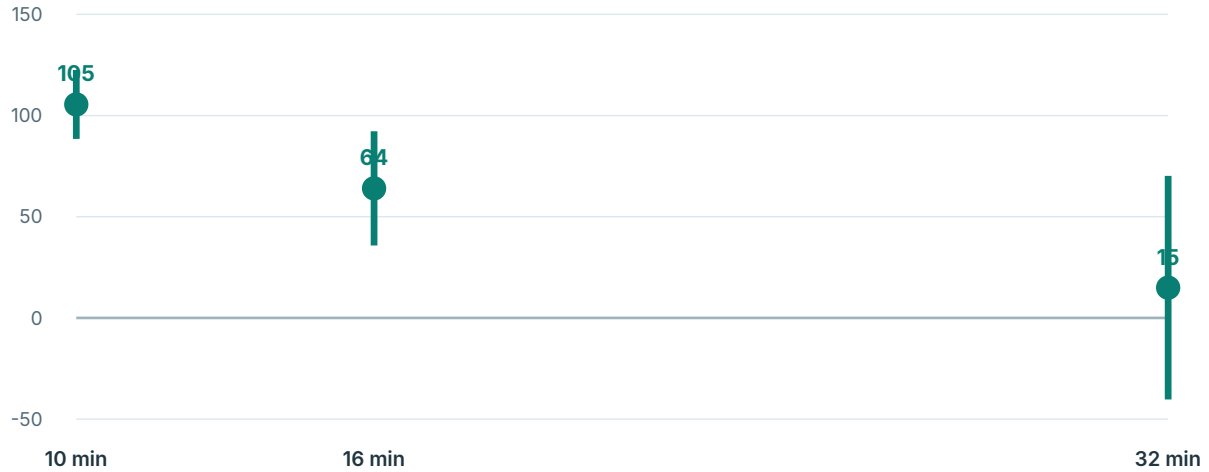
No green reallocation can serve all modeled straight demand. Mean end backlog increases from 54.4 vehicles at 10 min to 110.1 at 32 min.

INTERPRETATION

Use the recommendation within its scope

The timing advantage depends on the operating period

Input minus demand-proportional waiting; positive values favor 28/16/24/12 s



5,000 matched fresh runs per horizon, seed 2027; the 32-minute interval crosses zero.

10 min

Timing gain resolved

32 min

Difference unresolved

Paired 95% interval crosses zero: −40.3 to 70.2 vehicle-s.

Next: 90-second live demo · traffic-signal-analysis.pages.dev

Field counts and calibration are needed before real-world use.